

POWERED BY **ZAIZENS** · AN INTEGRATED HMS

PREMIUM · PROFESSIONAL EDITION

Laboratory & LIMS

Complete staff guide — payment validation, order alerts, structured templates, sample tracking, results registry, reports, corrections, and clinical integration.

PRODUCT

ZAIZENS Integrated HMS

DOCUMENT TYPE

Laboratory User Manual

WEBSITE

<https://www.zaizens.com/>

ORGANISATION

TSSF (Shisong Annex) SOA

Version **2.0**

Generated **14 June 2026 at 18:49**

Platform **Node.js HMS Edition**

Languages **English · Français**

Introduction

Welcome to the **ZAIZENS Integrated HMS Laboratory User Manual**. The Laboratory module manages the full diagnostic pathway — from doctor orders and cashier payment verification through sample collection, structured result entry, finalization, and patient report delivery.

Two workspaces: Use the **Laboratory Registry** (/laboratory) for results and the **LIMS hub** (/lims) for requests, samples, and configuration.

LAB
SERVICE CODES

LIMS
SAMPLE TRACKING

All
CLINICAL CONTEXTS



Table of contents

- 1. Laboratory overview & roles
- 2. End-to-end lab workflow
- 3. Laboratory portal
- 4. LIMS hub
- 5. Validate payment code
- 6. Order alerts
- 7. Lab templates & workbench
- 8. Laboratory registry
- 9. LIMS test requests
- 10. Sample collection
- 11. Entering & finalizing results
- 12. Reports & printing
- 13. Corrections & recalls
- 14. External lab results
- 15. OPD, IPD & Emergency orders
- 16. Cashier integration

17. Catalog & LIMS configuration

18. Screen reference

19. Troubleshooting

20. Glossary

1. Laboratory Overview & Roles

The **Laboratory module** connects clinical orders to verified, billable, and reportable test results across OPD, IPD, Emergency, and Maternity contexts.



Lab Technician

Validate LAB codes, enter structured results, collect samples, finalize reports



Doctor

Orders laboratory tests from consultation, IPD, or ER chart



Cashier

Collects payment; issues LAB service codes on order slips



Nurse / Phlebotomist

Sample collection, LIMS request workflow



Lab Supervisor

Review results, authorize corrections, manage LIMS config



Patient

Pays at cashier, presents LAB code at lab desk, receives printed report

Key URLs

Screen	Path
Laboratory Portal	/portal/lab
Laboratory Registry	/laboratory
Validate LAB code	/laboratory/validate
Lab templates / workbench	/laboratory/templates
Order alerts	/laboratory/order-alerts
LIMS hub	/lims
Lab requests	/lims/requests
Lab samples	/lims/samples
LIMS configuration	/lims/config
Test report	/laboratory/report/:id

Permissions: lab.read view orders & results · lab.write enter & validate results · service_catalog.laboratory.write catalog & LIMS config

2. End-to-End Lab Workflow

Doctor orders test → Patient pays at Cashier (LAB code issued)
→ Lab validates code → Sample collected (LIMS) → Results entered (templates)
→ Finalized → Report on chart & patient copy printed



Standard OPD pathway

- 1 Doctor adds laboratory test(s) during consultation.
- 2 Patient pays at Cashier — receives slip with **LAB-...** service code.
- 3 Lab technician validates code at `/laboratory/validate` .
- 4 Sample collected; LIMS request updated if used.
- 5 Technician opens lab template workbench and enters structured parameters.
- 6 Result finalized — appears on patient chart and Laboratory Registry.
- 7 Print patient copy from report or registry action menu.

Order alerts shortcut

Doctor orders also appear on **Order alerts** (`/laboratory/order-alerts`). Acknowledge alert → confirm → opens lab template with patient prefilled.

Payment gate: Lab cannot process unpaid orders. System blocks validation until Cashier marks the order line as paid.

3. Laboratory Portal

The **Laboratory Portal** (`/portal/lab`) is the lab technician home screen with KPIs and quick links.

KPI strip

- **Total Tests** — all registered lab results
- **Pending** — awaiting completion
- **Completed** — finalized results
- **Today's Tests** — tests registered today

Quick actions

- **LIMS hub** — requests, samples, configuration
- **Registry** — full results list at `/laboratory`
- **Templates** — structured result entry workbench
- **Order alerts** — new doctor orders needing attention
- **Patients** — patient search (if permitted)

Portal tiles provide role-specific shortcuts configured by your administrator.

4. LIMS Hub

The **Laboratory Information Management System (LIMS)** hub at `/lims` tracks the operational side of lab work — requests, samples, and configuration — complementing the results registry.

Hub KPIs

Stat	Meaning
Today's open	Requests scheduled today not yet completed
Submitted	New requests awaiting acceptance
In progress	Accepted or actively being processed
Samples collected	Physical samples recorded in system

LIMS menus

 Lab requests	 New request	 Lab samples	 Test results
 Validate code	 Templates	 Configuration	 Registry

Test results menu redirects to the Laboratory Registry (`/laboratory`) where finalized reports are managed.

5. Validate Payment Code

The validate screen is the primary entry point when a patient arrives at the lab desk with a payment slip.

- 1 Open /laboratory/validate or click + **New test** on the Registry.
- 2 Scan barcode or type the **LAB service code** (e.g. LAB-4829-K7HM3R9Q).
- 3 System verifies: code belongs to laboratory · order line is paid · ticket validity passes.
- 4 Validate detail shows patient, referring doctor, and in-house test lines.
- 5 Choose action: open lab template for structured entry, or submit quick findings.

Validation failures

Error	Cause
Payment required	Patient has not paid at Cashier
Code belongs to radiology/pharmacy	Wrong department code entered
Expired ticket	Consultation-linked validity expired
Duplicate result	Result already registered for this order line

Use the API preflight at validate time — the system checks authorization before opening templates to prevent unpaid processing.

6. Order Alerts

Order alerts (`/laboratory/order-alerts`) surface doctor-ordered tests from consultations, IPD, and Emergency contexts.

- 1 Open Order alerts from portal or sidebar.
- 2 Review **Needs your attention** — unacknowledged orders.
- 3 Each alert shows patient, test name, context (OPD / IPD / Emergency), and ordering doctor.
- 4 Click **Confirm & open lab template** — prefills patient and order line.
- 5 Or **Mark as seen** if handled through validate-code flow.

Recent history shows previously acknowledged alerts for audit trail.

Alerts for IPD and Emergency patients may appear before cashier settlement on credit tabs — verify local policy for when to process vs when payment is required.

7. Lab Templates & Workbench

The **Lab Templates** workbench (`/laboratory/templates`) provides structured data entry for common test panels — CBC, urinalysis, chemistry, serology, and custom templates.

Opening the workbench

- From validate screen after code verification
- From order alert confirmation
- Direct URL with `?code=LAB-...&oi=...` query parameters
- + **New test** on Laboratory Registry (with patient authorization)

Template entry

- 1 Select test template matching the ordered service.
- 2 Enter parameter values — numeric fields, dropdowns, reference ranges shown where configured.
- 3 Add findings and conclusion text for narrative sections.
- 4 Attach supporting documents if required.
- 5 Save draft or finalize — status moves to **RESULTS READY**.

Template definitions are maintained under `/laboratory/test-templates` (admin/lab supervisor).

8. Laboratory Registry

The **Laboratory Registry** (`/laboratory`) lists all test results with search, pagination, and actions.

Columns

Column	Description
Test ID	Internal reference <code>#LAB-0001</code>
Patient	Name and patient ID
Test name	Catalog or free-text test name
Referred by	Ordering doctor or self-referral
Date	Appointment / collection date
Status	Pending, In progress, Results ready, Completed, Revision

Status badges

PENDING

IN PROGRESS

RESULTS READY

RETURN FOR CORRECTION

Actions menu

- **View test** — open full report
- **Print patient copy** — when result is ready
- **Return for correction** — supervisor recall with audit note

9. LIMS Test Requests

Create and manage formal lab requests through LIMS when your facility uses request/sample tracking.

New request

- 1 Open `/lims/request/new` .
- 2 Select patient and prescribing doctor.
- 3 Choose collection center and scheduled date/time.
- 4 Select test group (panel) or individual catalog tests.
- 5 Submit — request status **Submitted**.

Request lifecycle

Status	Action
Submitted	Lab accepts request
Accepted	Mark in progress when work begins
In progress	Record samples, enter results
Done	Request completed — linked to registry result

Filter requests: Today · Pending · Done · All at `/lims/requests` .

10. Sample Collection

Physical sample tracking at `/lims/samples` and per-request **Samples** tab.

- 1 Open lab request detail page.
- 2 Go to **Samples** tab.
- 3 Record sample type (blood, urine, swab, etc.), collection time, and collector.
- 4 Sample status: collected → in transit → received in lab.
- 5 Sample list visible on LIMS samples page for batch review.

Sample types and collection centers are configured under LIMS Configuration.

11. Entering & Finalizing Results

Quick findings (validate flow)

On validate detail, enter findings and conclusion per order line and submit — creates registry entry with status **received**.

Structured templates

Preferred for panels with many parameters — ensures consistent formatting and reference range display on printed reports.

Finalization checklist

- All ordered lines have results entered
- Critical values flagged per local protocol
- Conclusion / interpretation documented
- Supervisor review if required by policy
- Status = Results ready or Completed

Results sync to patient chart — doctors view under Laboratory section of clinical record.

12. Reports & Printing

View report

`/laboratory/report/:id` — formatted report with facility header, patient demographics, test parameters, reference ranges, and conclusion.

Print patient copy

- From Registry action menu → Print patient copy
- From report page → `?print=1` query
- Diagnostic handover API for integrated print dialogs

Batch print

When multiple tests share one LAB service code, batch print API returns all finalized reports for patient handoff at once.

Reports use facility branding configured in deployment settings.

13. Corrections & Recalls

Return for correction

Supervisor with `lab.write` can recall a finalized result:

- 1

Registry → action menu → **Return for correction**.
- 2

Enter audit reason (required for compliance).
- 3

Status changes to **Return for correction** — revision pending flag set.
- 4

Re-open via validate code or templates to edit and resubmit.

Correction reason on resubmit

When editing a recalled result, document correction reason — stored in audit trail linked to order item.

Recalled results linked to billed orders reopen the validate-by-code path to maintain payment integrity.

14. External Lab Results

Tests performed outside the facility (external referral) are documented separately:

- Mark order line as **external** on validate screen
- Upload result document via **External upload** or **Attach document**
- Status shows as **External** on registry
- Document appears on patient chart as attached diagnostic report

Patient obtains test at external lab; staff uploads PDF or image when results return.

15. OPD, IPD & Emergency Orders

Context	Order source	Payment
OPD	Consultation orders	Prepay at Cashier before validate
IPD	Inpatient chart orders	Credit tab — settle at discharge or per policy
Emergency	ER chart orders	Retrospective billing after care
Maternity	ANC / labor orders	MAT tickets or standard cashier flow

Order alerts show context badge: Outpatient · Inpatient · Emergency — so technicians prioritize correctly.

IPD/Emergency alerts may route from ward board or ER disposition workflows.

16. Cashier Integration

Laboratory depends on Cashier for payment verification:

- 1

Doctor order creates OPD order line with pending payment status.
- 2

Cashier collects fee and marks line paid.
- 3

System generates

LAB-...

 service code on receipt.
- 4

Lab validate checks `isOrderLinePaidForFulfillment` and ticket validity.
- 5

Partial payments: patient returns to cashier until balance zero before lab proceeds.

See the **Cashier User Manual** (`/docs/cashier-users-manual`) for payment methods, wallets, and code validity rules.

17. Catalog & LIMS Configuration

Service catalog

Laboratory tests and prices maintained in Service Catalog → Laboratory. Import price lists via `/catalog/laboratory/import-price-list`.

LIMS configuration

`/lims/config` sections:

- **Test groups** — panels combining multiple catalog tests
- **Collection centers** — phlebotomy locations
- **Sample types** — blood, urine, CSF, etc.
- **Catalog linkage** — active tests available for requests

Requires `lab.write` or `service_catalog.laboratory.write`.

18. Screen Reference (UI)

Laboratory Portal — /portal/lab

Laboratory Portal

Validate codes · Enter results · Track samples

Total 128

Pending 12

Done 96

Today 8

LIMS hub

Registry

Templates

Order alerts

Validate code — /laboratory/validate

Laboratory · Validate code

LAB-4829-K7HM3R9Q

Jane Doe · Dr. Mbarga · CBC + Malaria RDT

✓ Paid · Open template

LIMS Hub — /lims

6

Today open

Requests

3

Submitted

New

2

In progress

Samples

5

Samples

Validate

Live screenshots: Open /portal/lab or /laboratory while signed in; Print → Save as PDF for training materials.

19. Troubleshooting

Issue	Solution
Code rejected — payment required	Patient must pay at Cashier first; verify order line status
Wrong department code	LAB codes start with LAB; RAD and PHA codes go to respective departments
Duplicate result warning	Result already exists for order line — view existing report or recall for correction
Template not loading	Confirm code and oi query params; check order alert context API
Cannot finalize result	Ensure all required template fields filled; check lab.write permission
Print blank report	Verify status is Results ready; try <code>?print=1</code> directly
Order alert missing	Doctor may not have saved order; check consultation chart orders tab
LIMS request stuck	Accept request → mark in progress → record sample → complete
External upload fails	Check file size limits; use PDF or image format
Expired consultation ticket	Patient needs new consultation payment if validity expired

20. Glossary

Term	Definition
LAB code	Service code prefix for laboratory order lines (e.g. LAB-4829-...)
LIMS	Laboratory Information Management System — requests & samples
Registry	Master list of all lab results at /laboratory
Validate	Verify payment and open order for processing
Template	Structured form for entering test parameters
Order alert	Notification of new doctor-ordered test
Order line (oi)	Single test item on OPD/IPD/ER order
Revision pending	Result recalled and awaiting correction
In-house	Test performed at this facility
External	Test performed outside; results uploaded
Panel / test group	Bundle of tests ordered together in LIMS